

Sample space and Events

Consider an experiment whose outcome is not predictable with certainty.

Suppose the set of all possible outcomes is known.

The set of all possible outcomes of an experiment is known as the sample space of the experiment and is denoted by S .

Eg:

① If the experiment consists of flipping two coins, then the sample space consists of

four points

$$S = \{ (H, H), (H, T), (T, H), (T, T) \}$$

② If an experiment consists of tossing three dice, then the sample space consists of

$$6^3 = 216 \text{ points}$$

$$S = \{ (a, b, c) \mid 1 \leq a, b, c \leq 6 \}$$

Here " a " is the number appear on the first die, " b " is the number appears on the second die and " c " is the number appear on

the third die.

③ If the experiment consists of measuring (in hours) the lifetime of a transistor, then the sample space consisting of all non-negative real numbers

$$S = \{x \mid 0 \leq x < \infty\}.$$

Event: Any subset E of the sample space S is known as an event.

If the outcome of the experiment is contained in E then we say that E has occurred

Eg: In an experiment of tossing three dice

$$E = \{ (1,1,1), (2,2,2), (3,3,3), (4,4,4), (5,5,5), (6,6,6) \}$$

is the event that same number appears on all 3 dice.

For any two events E and F of an experiment, we define a new event:

$E \cup F$: event that consists of outcomes that are either in E (or) in F .

The event $E \cup F$ occurs if either E or F occurs

$E \cap F$: event that contains all outcomes that are in both E and F .

Eg:

Suppose

$$E = \{ (H, H), (H, T), (T, H) \}$$

- event that atleast one head appears.

$$F = \{ (T, T), (T, H), (H, T) \}$$

- event that atleast one tail appears

$$E \cap F = \{ (H, T), (T, H) \}$$

event that exactly one head appears.

- If $E \cap F = \emptyset$ then the events E and F are called

"mutually exclusive".

- If E is an event then E^c denote the event that contains all possible outcomes that are not in E .

E^c occur iff E does not occur.

$$\boxed{E^c = S \setminus E}$$

Commutative Law

$$E \cap F = F \cap E$$

Associative Law

$$(E \cup F) \cup G \\ = E \cup (F \cup G)$$

Distributive Law

$$(E \cup F) \cap G$$

De Morgan's law:

$$\left(\bigcup_{i=1}^n E_i \right)^c = \bigcap_{i=1}^n E_i^c$$

and

$$\left(\bigcap_{i=1}^n E_i \right)^c = \bigcup_{i=1}^n E_i^c$$

Axioms of probability

Consider an experiment whose sample space is S . For each event E , we assume that a number $P(E)$ is defined and satisfy the following

three axioms.

Axiom 1

$$0 \leq P(E) \leq 1$$

Axiom 2

$$P(S) = 1.$$

Axiom 3

For any sequence of mutually exclusive events $E_1, E_2, \dots, E_n, \dots$ (i.e., $E_i \cap E_j = \emptyset$ for $i \neq j$)

$$P\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} P(E_n)$$

We call $P(E)$ as the probability of the event E .

Observation

- ① For any event E , we can write

$$S = E \cup E^c.$$

This implies that

$$1 = P(S) = P(E) + P(E^c)$$

and so,

$$P(E^c) = 1 - P(E)$$

- ② Since ϕ (empty set) is an event of S

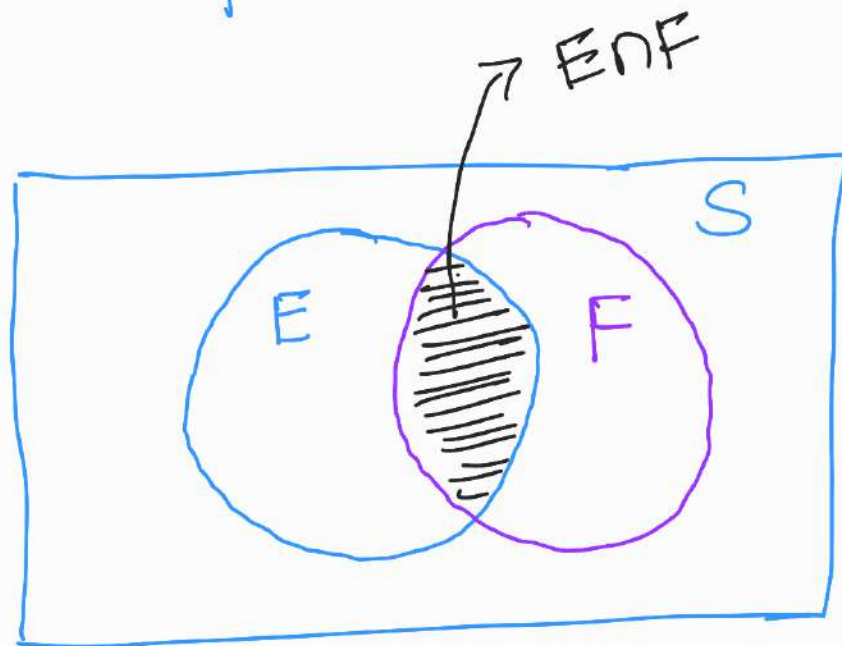
and $S = S \cup \phi$

$$p(S) = p(S) + p(\emptyset)$$

$$\Rightarrow 1 = 1 + p(\emptyset)$$

$$\Rightarrow \boxed{p(\emptyset) = 0}$$

③ Let E and F be events of an experiment



Note that $E \cup F$ can be written as

$$E \cup F = E \cup (F \cap E^c)$$

the union of two mutually exclusive events.

$$P(E \cup F) = P(E) + P(F \cap E^c) \longrightarrow \textcircled{1}$$

$$\text{Also, } F = (F \cap E^c) \cup (E \cap F)$$

then

$$P(F) = P(F \cap E^c) + P(E \cap F)$$

$$\Rightarrow P(F \cap E^c) = P(F) - P(E \cap F)$$

Substitute this in equation (1)

we get

$$P(E \cup F) = P(E) + P(F) - P(E \cap F)$$

Eg: 1 A person J carrying two books along during holiday trip. Here J likes first book with 0.5 probability, second book with 0.4 probability and both books with 0.3 probability.

What is the probability that J likes neither of books?

Solⁿ:- let B_1 - event that J likes the first book

B_2 - event that J likes the second book

$$P(B_1) = 0.5, \quad P(B_2) = 0.4$$

$$P(B_1 \cap B_2) = 0.3$$

$B_1^c \cap B_2^c$ - event that J likes neither of the books

By De Morgan's Law

$$B_1^c \cap B_2^c = (B_1 \cup B_2)^c$$

$$P(B_1^c \cap B_2^c)$$

$$= P((B_1 \cup B_2)^c)$$

$$= 1 - P(B_1 \cup B_2)$$

$$= 1 - [P(B_1) + P(B_2) - P(B_1 \cap B_2)]$$

$$= 1 - [0.5 + 0.4 - 0.3]$$

$$= 1 - 0.6$$

$$= 0.4.$$

Sample space having equally
likely events

Consider an experiment
whose sample space S is
a finite set.

If the probability of all
outcomes are same then
it is called "equally likely
outcomes".

In this case, for any event
 E , the probability

$$p(E) = \frac{n(E)}{n(S)},$$

$n(E)$ - number of elements
in the set E

$n(S)$ - number of elements
in the sample space

Eg: ① If 3 balls are

drawn randomly from a
bowl containing 6 white
and 5 black balls.

What is the probability

that One of the ball
is white and the other
two are black ?.

Solⁿ: Case 1:

If we consider that balls
are being drawn are
distinguishable and so,
order of selection is relevant.

The total number of ordered
Selection (ordered arrangement)
of 3 ball from 11 balls

is,

$$11 \times 10 \times 9 = 990$$

Sample space S has 990
out comes

$$n(S) = 990$$

Let E be the event that
one of the ball is white
and the other two are
black.

E_1 - event that first ball
is white other
two are black

E_2 - event that second ball is white other two are black.

E_3 - event that third ball is white other two are black.

$$P(E_1) = \frac{6 \times 5 \times 4}{990} = \frac{120}{990}$$

$$P(E_2) = \frac{5 \times 6 \times 4}{990} = \frac{120}{990}$$

$$P(E_3) = \frac{5 \times 4 \times 6}{990} = \frac{120}{990}$$

$$\text{Since } E = E_1 \cup E_2 \cup E_3$$

$$\begin{aligned}
 P(E) &= P(E_1) + P(E_2) + P(E_3) \\
 &= \frac{360}{990} \\
 &= \frac{4}{11}.
 \end{aligned}$$

Case ②: This problem could be solved by unordered set of drawn balls.

The sample space S has

$$\binom{11}{3} = \frac{11!}{8! \cdot 3!} = 165$$

out comes

$$n(S) = 165$$

E - event that one of the ball is white and other two balls are black.

$$n(E) = \binom{6}{1} \binom{5}{2} \\ = 60$$

$$P(E) = \frac{n(E)}{n(S)} \\ = \frac{60}{165} \\ = \frac{4}{11}.$$

Remark: When the experiment consists of random selection of k items from a set of n items, we have the flexibility of either letting the outcome of the experiment be ordered selection of k items (or) letting it be unordered selection.

For instance: ② Suppose 5 people are to be randomly selected from a group of 20 individuals

Consisting of 10 married couple.
What is the probability that
5 chosen are unrelated?

Solⁿ:- ^{Case 1} If we regard the
sample space as the set of
5 people chosen, then
there are $\binom{20}{5}$ equally
likely outcomes.

$$n(s) = \frac{20!}{15! \cdot 5!}$$
$$= 15,504.$$

E - event that 5 people chosen are unrelated.

An outcome that does not contain a married couple can be thought of as a result of six stage experiment.

In the first stage, 5 of the 10 couples to have a member in each group are chosen and next 5 stages 1 of the two members of each of the

Couple selected.

So,

$$n(E) = \binom{10}{5} 2^5$$

$$P(E) = \frac{n(E)}{n(S)}$$

$$= \frac{\binom{10}{5} 2^5}{\binom{20}{5}}$$

$$= 0.5201$$

Case 2

Now if we let the outcome of the experiment be the ordered selection of 5 individuals

$$n(S) = 20 \times 19 \times 18 \times 17 \times 16$$

E - event that 5 chosen people are unrelated.

$$n(E) = 20 \times 18 \times 16 \times 14 \times 12$$

$$\begin{aligned} \therefore P(E) &= \frac{\cancel{20} \times \cancel{19} \times \cancel{18} \times 14 \times 12}{\cancel{20} \times 19 \times \cancel{18} \times 17 \times \cancel{16}} \\ &= 0.5201. \end{aligned}$$

③ A Committee of 5 u's to be selected from a group of 6 men and 9 women. If the selection u's made randomly, what u's the probability that the Committee consists of 3 men and 2 women?

Solⁿ:- A Committee of 5 members must be formed from 15 people, so the sample space S consists of $\binom{15}{5}$ outcomes

$$n(S) = \binom{15}{5} = 1001$$

E - event that the Committee consists of 3 men and 2 women.

$$n(E) = \binom{6}{3} \binom{9}{2} = 240$$

The desired probability

$$\begin{aligned} p(E) &= \frac{n(E)}{n(S)} \\ &= \frac{240}{1001} \end{aligned}$$

④ An urn consists n balls, one of which is special.

If k of these balls are withdrawn one at a time, with each selection being equally likely to be any of the balls that remain at that time, what is the probability that the special ball is chosen?

Solⁿ:- Since all balls are treated in an identical manner, it follows that the set of k balls selected is equally

likely to be any of $\binom{n}{k}$ sets
of k balls.

$$n(S) = \binom{n}{k}$$

E - event that the special
ball is chosen in that
selection of k balls.

$$n(E) = \binom{1}{1} \binom{n-1}{k-1}$$

$$P(E) = \frac{n(E)}{n(S)} = \frac{\binom{n-1}{k-1}}{\binom{n}{k}}$$

$$= \frac{(n-1)!}{(\cancel{n-k})! (k-1)!} \frac{(\cancel{n-k})! k!}{n!}$$

$$= \frac{k}{n}$$

(OR)

This can also be solved by

letting

A_i - event that i^{th}
ball chosen is special
one

$$i=1, 2, \dots, k$$

Since there are n balls
and i is a sample space
of equally likely outcomes,

$$p(A_i) = \frac{1}{n} \quad \forall i=1,2,\dots,k$$

Moreover, $E = A_1 \cup A_2 \cup \dots \cup A_k$

$$p(E) = p(A_1) + p(A_2) + \dots + p(A_k)$$

$$= \frac{1}{n} + \frac{1}{n} + \dots + \frac{1}{n}$$

$$= \frac{k}{n}$$

why $p(A_i) = \frac{1}{n}$?

note that there are

$$n(n-1)(n-2) \dots (n-k+1)$$

$$= \frac{n!}{(n-k)!}$$

ways of ordered selection of k -balls.

For event A_i , the i^{th} ball is the special ball.

There are.

$$\begin{array}{ccccccc} (n-1) & (n-2) & \dots & (n-i+1) & \cdot & 1 & (n-i) \dots \dots \\ & \uparrow & & \uparrow & & \uparrow & \uparrow \\ & 1^{\text{st}} \text{ ball} & & 2^{\text{nd}} \text{ ball} & & (i-1)^{\text{th}} \text{ ball} & i^{\text{th}} \text{ ball} \\ & & & & & & \text{special} \\ & & & & & & (n-k+1) \end{array}$$

$$= \frac{(n-1)!}{(n-k)!}$$

The probability that the i^{th} ball is special is

$$p(A_i) = \frac{(n-1)!}{(n-1)!} \cdot \frac{n!}{(n-1)!} = \frac{1}{n}$$

A deck of 52 cards

A pack of 52 cards

are 4 types

Cards (52)

spades(13)	clubs(13)	Diamonds(13)	Hearts(13)
K	K	K	K
Q	Q	Q	Q
J	J	J	J
A	A	A	A
2-10 cards	2-10 cards	2-10 cards	2-10 cards

② In a game of cards,
the entire deck of 52 cards
dealt evenly to 4 players.
What is the probability that

① One of the players receives
all 13 spades?

⑥ each player receives 13 cards?

Solⁿ. Here the sample space consists of all possible outcomes when 52 cards distributed evenly among 4 players.

$$n(S) = \binom{52}{13 \ 13 \ 13 \ 13}$$
$$= \frac{52!}{(13!)^4}$$

⑦ E - event that one of the player receives all 13 spades

If one player gets all 13 spades then the rest of the 3 players get even distribution of remaining 39 cards. Further, that one player receiving all 13 spades could be any one of 4 players.

$$\begin{aligned} n(E) &= \binom{4}{1} \binom{39}{13 \ 13 \ 13} \\ &= \frac{4 (39!)}{(13!)^3} \end{aligned}$$

$$P(E) = \frac{n(E)}{n(S)}$$

$$= \frac{4(39!)}{(13!)^3} \times \frac{(13!)^4}{52!}$$

$$= 6.3 \times 10^{-12}$$

⑥ let F be an event that each player receives 1 ace.

First distribute 4 aces among 4 players evenly and then distribute 48 cards evenly

among 4 plays. By applying basic principle of counting, we get

$$n(F) = \binom{4}{1 \ 1 \ 1 \ 1} \binom{48}{12 \ 12 \ 12 \ 12}$$
$$= 4! \cdot \frac{48!}{(12!)^4}$$

$$P(F) = \frac{n(F)}{n(S)}$$
$$= \frac{4! (48!)}{(12!)^4} \times \frac{(13!)^4}{52!}$$

$$= 0.1055.$$

③ If n people present in a room then what is the probability that no two of them celebrate their birthday on the same day of the year?

How large this ' n ' should be so that the probability is less than $1/2$?

Solⁿ: As each person can celebrate his (or) her birthday on any

one of 365 days, there are a total of $(365)^n$ possible outcomes

E - event that no two celebrate birthday on the same day.

$$n(E) = 365 \times 364 \times \dots \times (365 - n + 1)$$

$$P(E) = \frac{n(E)}{n(S)}$$

$$= \frac{(365) \cdot (364) \cdot \dots \cdot (365 - n + 1)}{(365)^n}$$

Here $n \geq 23$ so that the desired probability is less than $\frac{1}{2}$.

④ A total of 36 members of a club play tennis, 28 play squash and 18 play badminton. Furthermore 22 of the members play both tennis and squash, 12 play both tennis and badminton, 9 play both squash and badminton and 4 play all the sports. How many members of the club play at least one of the three sports?

Solution: Let N be total number of members in the club.

T - event that members of club
playing tennis

S - event that members of club
playing squash

B - event that members of club
playing badminton.

$$P(T) = \frac{n(T)}{N} = \frac{36}{N}$$

$$P(S) = \frac{n(S)}{N} = \frac{28}{N}$$

$$P(B) = \frac{n(B)}{N} = \frac{18}{N}$$

$$P(Tns) = \frac{n(Tns)}{N} = \frac{22}{N}$$

$$p(TnB) = \frac{n(TnB)}{N} = \frac{12}{N}$$

$$p(snB) = \frac{n(snB)}{N} = \frac{9}{N}$$

$$p(TnsnB) = \frac{n(TnsnB)}{N}$$

$$= \frac{4}{N}$$

The desired probability

$$P(TUSUB)$$

$$= P(T) + P(SUB) - P(Tn(SUB))$$

$$= p(T) + p(S) + p(B) - p(S \cap B) \\ - p((T \cap S) \cup (T \cap B))$$

$$= p(T) + p(S) + p(B) - p(S \cap B) \\ - p(T \cap S) - p(T \cap B) \\ + p(T \cap S \cap B)$$

$$= \frac{36}{N} + \frac{28}{N} + \frac{18}{N} - \frac{9}{N} - \frac{12}{N} \\ - \frac{22}{N} + \frac{4}{N}$$

$$= \frac{43}{N}$$

This means that

$$P(TUSUB) = \frac{n(TUSUB)}{N}$$

$$= \frac{43}{N}.$$

Equivalently,

$$n(TUSUB) = 43.$$

We conclude that 43 members play at least one of the sports.

Note that we computed the probability of the event $TUSUB$ as the total number of members of club is not given.

Ex: Suppose $E_1, E_2, E_3, \dots, E_N$
are events of an experiment

then

$$P\left(\bigcup_{i=1}^N E_i\right)$$

$$= \sum_{i=1}^N P(E_i) - \sum_{i_1 < i_2} P(E_{i_1} \cap E_{i_2})$$

$$+ \dots + (-1)^{n+1} \sum_{i_1 < i_2 < \dots < i_n} P(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_n})$$

$$+ \dots + (-1)^{N+1} P(E_1 \cap \dots \cap E_N)$$